

Where Is the Agent Socially Deficient? Low-Rank Skill Sensing for LLM Agents in a Generative Social Sandbox

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1. Motivation

Social abilities such as small talk, negotiation and leadership have no rule-based simulator and little real training data, and reproducing the human data-generating process is infeasible. What we *can* get is a small evaluation set of social tasks with a verifier: SOTOPIA-style goal-completion scores [1] or human ratings. Everyday social life is also highly compressible: most of it follows cultural scripts (ordering, greeting, a class), and individual differences show up at a few choice points [2]. LLM-based synthetic populations are known to answer too consistently and to distort relationships between traits [3]. We turn these observations into an ML problem: **learn a low-dimensional map of which social skills a scenario demands and which ones an agent lacks, from a few (agent, scenario) scores.**

2. The social sandbox as an underspecified POMDP

A scenario $\theta = (\sigma, \xi, g_{1:n})$ has three parts:

- a **script graph** σ , whose nodes are options (roles, stages, norms, exits) [4];
- a **perturbation** ξ , either a break (something disrupts the script) or an opportunity (a cue that can trigger a new goal);
- **cognitive genomes** $g_j \in [0, 1]^p$ for the non-player characters. Each coordinate is one mechanism, such as loss aversion, conformity or social anxiety, calibrated on behavioural assays (Psych-101, choices13k). It is expressed in a frozen LLM through a prompt template or a steering vector, $h_\ell \leftarrow h_\ell + \sum_c g_{jc} v_c$.

The agent π_w is an LLM with a LoRA adapter w . It observes the dialogue and scene, acts with utterances and moves, and is scored by a verifier $J(\tau) \in [0, 10]$. We release the sandbox as a Gym-style multi-agent text environment. The pixel-art town demo is its visual front end, and it currently runs on hand-set rules, not on trained models.

3. Learning problem

Regret matrix. A population of N agents (checkpoints, base models, prompting strategies) is scored on M scenarios, with $R_{ij} = J(\pi_{\text{ref}}, \theta_j) - J(\pi_i, \theta_j)$. Only a sparse set of entries Ω is observed, because each entry costs a full multi-turn rollout.

Inductive matrix completion with sparse deficiencies. We learn a scenario encoder $\phi_\omega(\theta) = f_\omega(\text{text}(\sigma, \xi), g_{1:n}) \in \mathbb{R}^d$ and agent deficiency vectors u_i :

$$\min_{\omega, U} \sum_{(i,j) \in \Omega} (R_{ij} - u_i^\top \phi_\omega(\theta_j))^2 + \lambda \|U\|_F^2 + \mu \sum_i \|u_i\|_1.$$

Side information (the text and genomes) lets the model generalize to unseen scenarios, which plain completion [5] cannot do. For a *new* agent we probe n scenarios and recover its deficiency with a LASSO; the RL companion project analyses $n = O(k \log(d/k))$ under RIP-type conditions.

Targeted scenario generation. An LLM scenario writer is conditioned on a target demand vector $\phi^\dagger \propto \hat{u}$. Its outputs pass an **admission classifier**, trained to reject implausible scenarios, and are ranked by predicted regret $\phi_\omega(\theta)^\top \hat{u}$. The agent is then fine-tuned on the admitted scenarios.

Critical-turn detection (temporal compression). For logged dialogues we measure counterfactual turn influence, $\Delta_t = \mathbb{E}[J \mid \text{resample turn } t] - J$, and train a cheap classifier to predict high $|\Delta_t|$ from context. Training focuses on the predicted critical turns, and routine script segments reuse cached transcripts instead of fresh rollouts.

4. Evaluation

Data	SOTOPIA scenarios and its hard subset; our genome-parameterized NPC partners; behavioural assays for calibrating the genomes.
Models	Qwen2.5-1.5B/7B-Instruct with LoRA as agents; a frozen LLM with steering vectors for NPCs.
Baselines	Uniformly random scenarios (domain randomization); SOTOPIA- π -style behaviour cloning plus self-reinforcement [6]; PLR-style replay of high-score-gap scenarios.
Metrics	Goal completion on held-out hard scenarios; RMSE of held-out entries of R vs. $ \Omega $; number of probes needed to recover $\hat{u}$; canonical correlation between learned axes and genome dimensions; training compute saved by critical-turn selection; spread of NPC behaviour vs. a single-persona baseline.

Q1 Is R approximately low-rank for social tasks? **Q2** Do the learned skill axes line up with cognitive mechanisms, making skills readable and editable? **Q3** Does targeted generation beat random and replay baselines at equal rollout budget? **Risk:** verifier noise can inflate the rank of R . We average several verifier samples and report agreement with human ratings on a subset.

5. Plan

Wk 1–3 text sandbox API, scripts, genome expression for 4–6 mechanisms; **4–6** agent population, sparse score matrix, test Q1; **7–10** inductive completion, admission classifier, targeted generation, critical-turn detection; **11–13** held-out evaluation, write-up, demo. **Joint paper:** the RL project supplies the method and theory (MiniGrid, Craftax); this project supplies the social application.

[1] Zhou et al. SOTOPIA: Interactive Evaluation for Social Intelligence in Language Agents. ICLR 2024. [2] Schank, Abelson. Scripts, Plans, Goals, and Understanding. 1977. [3] Chidichimo, Jung, Wallis, He. Artificial Societies Benchmark: A Validation Framework for Synthetic Research. arXiv 2609.30030, 2026. [4] Sutton, Precup, Singh. Between MDPs and semi-MDPs. AIJ 1999. [5] Candès, Recht. Exact matrix completion via convex optimization. FoCM 2009. [6] Wang et al. SOTOPIA- π : Interactive Learning of Socially Intelligent Language Agents. ACL 2024. [7] Park et al. Generative Agents: Interactive Simulacra of Human Behavior. UIST 2023. [8] Dennis et al. PAIRED. NeurIPS 2020.